Preliminary Assignment Summary

Our group's analysis was precisely focused on getting information about each type of cancerous skin lesion. In this summary we will discuss about:

- Quality of the data
- Manual picture analysis
- Metadata analysis
- Annotation agreement

1. Quality of the data

Quality of the images:

Blurriness:

Some images are more blurred than others. Most of them have an acceptable level of blur, and their analysis will probably be accurate. However, some images may be too blurred or out of focus, which could lead a detection program to produce incorrect results. Below are examples of clear images (PAT_75_113_165.png) and examples of blurry or out-of-focus images (PAT_923_1753_496.png; PAT_73_111_609.png ; PAT_42_58_13.png; PAT_101_1041_898.png)

Light/colour grade:

Some images appear darker or lighter than others, not because of the actual color of the lesion, but due to differences in lighting exposure (PAT_115_177_622.png, PAT_134_201_587.png). Since the color of the lesion can be used during image analysis, incorrect or non-representative lighting can lead to false diagnoses and make comparisons between images more difficult.

Mask:

Some masks do not correspond accurately the lesion on the picture. This is caused by two reasons. The first one is that some lesions are very homogenous and with low contrast, that causes the mask to be inaccurate, here with is an example (PAT_1018_107_68.png).

The second reason is that there are some pictures of curved/intricate body parts that can causes shade, this also makes the mask inaccurate, here with is an example (PAT_87_133_391.png).

Metadata:

The flaw of the metadata is that there are some missing values/information regarding some personal information about the patients (smoking, drinking, origin of parents, gender, etc.). The absence of this information can mislead the diagnosis of each patient and most importantly the analysis of the whole sample group will probably give inaccurate statistic.

Our hypothesis:

After reviewing the dataset, we observed that every patient with missing information has a benign skin disease. We conclude that since these cases are not associated with skin cancer condition, the missing data is likely less relevant for diagnosis, which could explain why this information was not systematically collected.

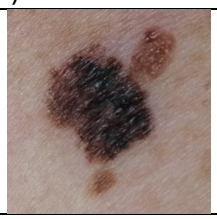
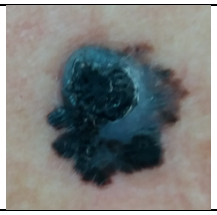
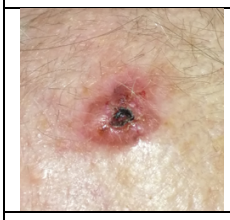
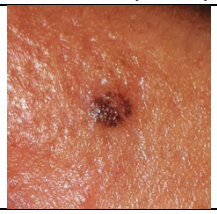
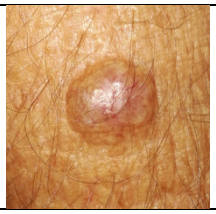
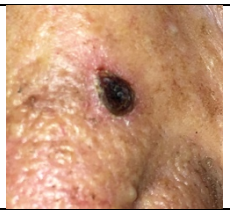
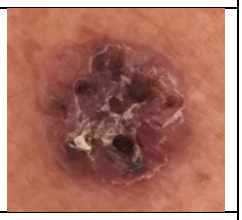
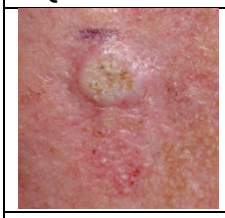
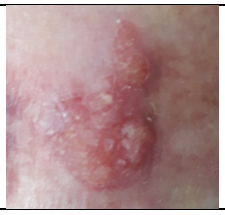
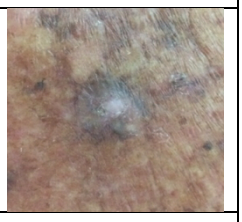
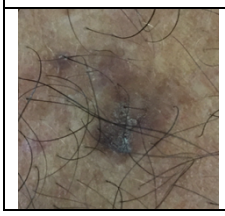
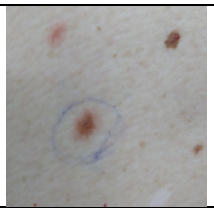
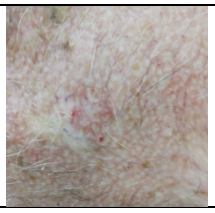
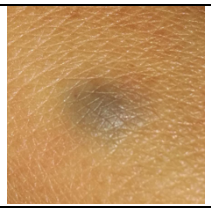
metadata

	Unnamed: 0	patient_id	lesion_id	smoke	drink	background_father	background_mother	age	pesticide	gender	skin_cancer_history	cancer_history	has_piped_water	has_sewage_system	fitspatrick	region	diameter_1	diameter_2	diagnostic	itch	grew	hurt	changed	bleed	elevation	img_id	biopsed	group_id
25	26	PAT_1902	3790					66								FOREARM			SEK	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	PAT_1902_3790_168.png	FALSE	F
26	27	PAT_42	58	FALSE	FALSE	GERMANY	GERMANY	38	FALSE	FEMALE	TRUE	TRUE	TRUE	TRUE	1.0	CHEST	10.0	9.0	BCC	TRUE	UNK	TRUE	UNK	TRUE	FALSE	PAT_42_58_13.png	TRUE	F
30	31	PAT_134	201	FALSE	FALSE	POMERANIA	POMERANIA	62	FALSE	MALE	TRUE	FALSE	FALSE	FALSE	1.0	NOSE	10.0	6.0	BCC	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	PAT_134_201_587.png	TRUE	F
39	40	PAT_793	1512	FALSE	FALSE	ITALY	ITALY	36	FALSE	MALE	FALSE	TRUE	TRUE	FALSE	4.0	BACK	7.0	4.0	NEV	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	PAT_793_1512_327.png	TRUE	F
41	42	PAT_1842	3615					17								FOREARM			NEV	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	PAT_1842_3615_850.png	FALSE	F
55	56	PAT_183	283	FALSE	FALSE	UNK	UNK	76	FALSE	MALE	TRUE	FALSE	FALSE	FALSE	3.0	CHEST	9.0	7.0	SCC	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	PAT_183_283_495.png	TRUE	F
100	102	PAT_1573	2499					45								FACE			ACK	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	PAT_1573_2499_849.png	FALSE	F
114	117	PAT_875	1669	FALSE	FALSE	POMERANIA	POMERANIA	61	TRUE	MALE	TRUE	FALSE	TRUE	TRUE	2.0	BACK	19.0	11.0	BCC	TRUE	UNK	FALSE	UNK	TRUE	TRUE	PAT_875_1669_946.png	TRUE	F
123	127	PAT_999	20					40								ARM			ACK	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	PAT_999_20_223.png	FALSE	F
145	149	PAT_875	1670	FALSE	FALSE	POMERANIA	POMERANIA	61	TRUE	MALE	TRUE	FALSE	TRUE	TRUE	2.0	BACK	9.0	8.0	BCC	TRUE	UNK	FALSE	UNK	FALSE	TRUE	PAT_875_1670_121.png	TRUE	F
161	165	PAT_396	796	FALSE	TRUE	PORTUGAL	ITALY	76	FALSE	FEMALE	FALSE	FALSE	FALSE	FALSE	2.0	FACE	7.0	6.0	BCC	FALSE	TRUE	FALSE	FALSE	FALSE	TRUE	PAT_396_796_394.png	TRUE	F
172	176	PAT_1702	3548					57								BACK			NEV	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	PAT_1702_3548_386.png	FALSE	F
212	217	PAT_782	1493	FALSE	FALSE	GERMANY	GERMANY	71	FALSE	MALE	TRUE	TRUE	FALSE	TRUE	3.0	CHEST	12.0	6.0	ACK	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	PAT_782_1493_140.png	TRUE	F
233	238	PAT_948	1801	TRUE	FALSE	POMERANIA	POMERANIA	58	TRUE	FEMALE	FALSE	FALSE	FALSE	FALSE	3.0	NOSE	3.0	2.0	BCC	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	PAT_948_1801_993.png	TRUE	F
247	252	PAT_1333	1177					38								BACK			NEV	FALSE	TRUE	FALSE	FALSE	FALSE	TRUE	PAT_1333_1177_261.png	FALSE	F
257	264	PAT_1583	2607					71								FACE			SEK	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	PAT_1583_2607_628.png	FALSE	F

2. Manual picture analysis

We analysed the pictures manually and concluded the following features for each diagnosis:

- MEL: Very dark and intense pigment with rough and asymmetrical shape, often not crust
- BCC: Round shape with a black dot in the centre, often has relief and visible blood vessels.
- SCC: Often skin shed and scar-like shaped, mainly skin-coloured

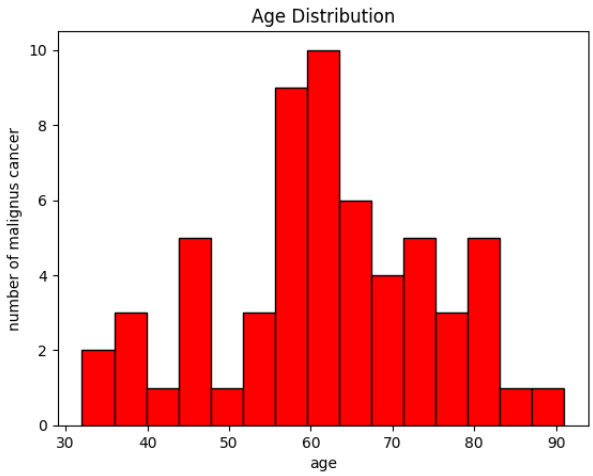
MELANOMA (MEL)				
				
BASAL CELL CARCINOMA (BCC)				
				
SQUAMOUS CELL CARCINOMA (SCC)				
				
BENIGN SKIN DISEASES				
				

3. Metadata analysis

Analysing the patient's data provided by the metadata.csv file, we came up with some very interesting relations between different factors, which may be connected to the current conditions of the patients in our sample group.

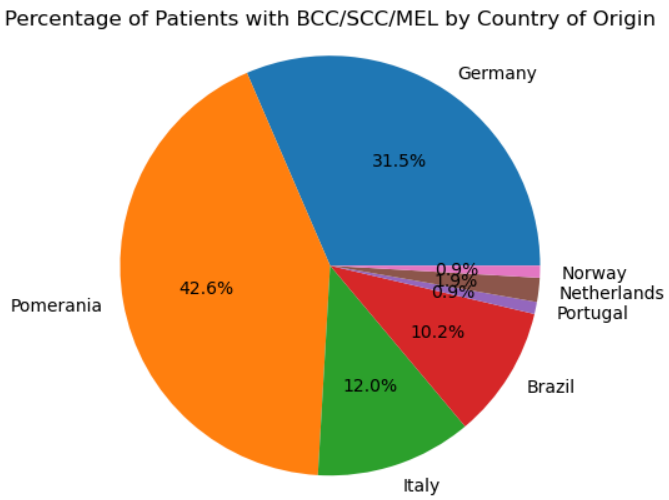
Age:

We can start by analysing the age of the patient so that we can check if the age affects the overall growth of such cells. From our sample data the median age of the patients with cancer diagnosis is as followed: BCC – 60.78 years, SCC – 67.80 years, MEL – 55.66 years, which means that overall people over the age of 50 are more susceptible on getting skin cancer than younger ones.



Geographical origin:

Regarding the countries or regions of origin, we also have some very interesting information. 74.1% of all the patients that have skin cancer have a father or mother of Northern/Northern-Eastern European origin (Germany and Pomerania region). So, 7 out of 10 people with cancer have origin from that part of Europe.



Drinking and smoking:

From the metadata we can see that exactly 90% of the active smokers have any form of skin cancer, so regarding our sample, we can conclude that 9 out of 10 smokers have cancer. This information may direct us to the conclusion that smoking may lead with high rate to developing skin cancer.

As for the smoking analysis, the results for drinking are also quite high – 76% of the drinkers have any form of skin cancer. So drinking alcohol may also be a factor of the development of the skin cancers.

These statistics are unfortunately not accurate because of the missing information. We calculated them excluding the patient who didn't provide any information about their drinking and smoking habits.

4. Our annotation agreement

We analysed our annotation and we arrived at the following conclusion.

Pen annotation:

For pen mark's annotation we agreed for most of them. We got 0.90 as Fleiss Kappa value. This is probably caused by the duality of the choice signed in the csv file as Boolean value 0 and 1.

Hair annotation:

For the hair's annotation we agreed partially, with a Fleiss Kappa value of 0.45. In this case the choice was between 0, 1, 2 or 3 so it is a greater range of options and gives more space to nuance. We also observed even if most of the annotation are quite similar (with discrepancies of 1 value), some of them are drastically opposite.

As an example of that we can use image PAT_822_1550_132.png. In this picture we can clearly see hair, but they do not cover the lesion. For this reason, two of us chose 0 for this annotation and other three chose 3. This shows that it is important to clearly state the parameters of the annotation or else misunderstanding can lead to this type of disagreement.

Collaborators (name surname/github username) : Valentina Grazzini / valegrazzini, Nikolay Valkov / Nikolayy06, Oscar-James L'huillier-Pedersen / Oscar-James, Julius Kerff Michelsen / JuliusMichelsen, Hafsa Ismail / Hais1-jpg

GitHub link: https://github.com/Oscar-James/ProjectInDataScience2026_ExamTemplate_F.git